

Katherine Titterton

Imaging, signal & ML scientist-engineer · neuroscience

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MACROFLUIDIC.ME/RESUME

Neuroscientist-engineer at the seam of imaging, signal extraction, and machine learning. Eight years building the instruments, datasets and models that turn biological signal into measurement, from voltage-clamp recordings and expansion microscopy to PyTorch segmentation over 100,000+ compound image sets.

100,000+

compounds imaged for the largest Cell Painting datasets

1+ year

my PyTorch segmentation models have run in production

~10×

more sensitive bioactivity detection than standard assays

8 papers

peer-reviewed, including two in Nature Communications

EXPERIENCE

[n] = publication number on page 2

- Axiom Bio** · Founding Member of the Technical Staff San Francisco, CA 2023–present
- Owned image labeling and segmentation: PyTorch U-Net models over 100,000+ compound Cell Painting images, in production 1+ year.
 - Owned bioactivity algorithm development and testing: invented two image-based methods ~10× more sensitive than conventional assays.
 - Oversaw assay development, QC and data generation for two of the world's largest Cell Painting datasets; drove product direction.
- Nat Commun 2026 [4] · Cell Syst 2026 [3]
- Instituto** · Associate Scientist III, Arrayed Screening, Neuroscience South San Francisco, CA 2022–2023
- Onboarded two neuroscience screens (TSC, ALS) onto automated platforms; imaged hundreds of 384-well plates, analyzed in Python.
 - Trained two screen operators; designed and validated a label-free phase-contrast timecourse imaging modality to track neuron growth.
- MIT, Synthetic Neurobiology Group** · Technical Associate II Cambridge, MA 2021–2022
- Imaged 30+ proteins at nanoscale by optimizing multiplexed expansion revealing (mExR) for 7+ IHC rounds on 16× expanded tissue.
 - Built and ran the image-analysis pipelines (ImageJ/Fiji, MATLAB) behind the published method.
- Nat Commun 2024 [6]
- AbbVie, Foundational Neuroscience Center** · Associate Scientist II Cambridge, MA 2018–2021
- Built primary murine and hiPSC neuron tau-aggregation high-content assays (Revvity Opera, Harmony) to rank Alzheimer's lead candidates.
 - Designed microfluidic neuron-culture devices to model tau propagation; authored R pipelines for reproducible dose-response profiling.
- Mol Neurodegener Adv 2026 [1] · iScience 2026 [2] · J Biol Chem 2025 [5] · iScience 2022 [7]
- Massachusetts General Hospital** · Research Technician I Boston, MA 2017–2018
- Recorded and analyzed voltage-clamp currents from GABA-A-receptor *Xenopus* oocytes to resolve mechanisms of general anesthetics.
- Eur J Med Chem 2019 [8]

PROJECTS

- Bare Metal** · Solo generative-art exhibit, Human Tech Week, SF; concept, fabrication and post-hoc analysis (macrofluidic.me/blog) 2025
- Vivarium 2.0** · Designed and built an automated poison-dart-frog habitat: misting control, motion-triggered live video 2024
- OpenTransformers** · LLM code-generation companion that lets wet-lab scientists drive the OT-2 robot; 2nd place, BioHack NYC 2023
- Open source** · github.com/kat-titter: vanitas-veritas, diffusion video pipeline; claude-code-learns, reusable agent skills from corrections 2026

EDUCATION

- Harvard Extension School** · Graduate coursework (Python; image processing & computer vision; social medicine), while employed full-time 2017–2020
- B.A., Colgate University** · cum laude 2013–2017
- Cellular Neuroscience major, Mathematical Systems Biology minor; biophysics, neuroethology. Co-captain, Division I Track & Field.

SKILLS & TOOLS

ML & DATA

PyTorch · U-Net segmentation (SAM evaluated) · image embeddings & clustering · Python (NumPy, SciPy, scikit-learn, pandas) · multivariate statistics · LLM codegen (Claude Code, Cursor)

HARDWARE & AUTOMATION

Liquid handlers & schedulers (OpenTrons, Integra, BioTek, Labcyte Echo, High-Res, Green Button Go) · microfluidic devices · SOP-driven validation & QC

IMAGING & SIGNAL

High-content & confocal microscopy (Opera Phenix, Operetta) · expansion microscopy · phase-contrast timecourse · voltage-clamp electrophysiology · ImageJ/Fiji, MATLAB

NEURO WET LAB

Primary murine & hiPSC neuron culture · ICC/IHC · stereotaxic surgery & microdissection · assay development & HT/HC screening

CERTIFICATIONS

FII Level 1 Freediving (Oct 2019) · Molchanov L2 Freediving (Oct 2025) · SII Shark Ecology Specialty (Apr 2020)

PEER-REVIEWED PUBLICATIONS

8 papers · newest first

- [1] Yanamandra K, Romanul N, Patel H, Sanders H, Chang R, Darcy A, Brown N, Kane C, Wu HY, Jagtap K, Chancellor S, Morozova O, Xi M, Hogan C, **Titterton K**, Wu JW, Manos J, Karran E, Dellovade T, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6 ABBVIE
- [2] Sebastian RM, Kupp R, Nanavati D, Parfentev I, Krzystek TJ, Romanul N, Yanamandra K, Preiss CN, Manos JD, Wu HY, Volkova A, Crescenti S, **Titterton K**, Welker AM, Townsend M, Karran E, Langlois X, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 PMID 42325569 · PMC13276770 ABBVIE
- [3] Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, Boiko DA, Cabrera AA, Cheah J, Cimini BA, Gorissen BL, Harrill J, Jones TR, Karczewski KJ, Crute CE, Rouquie D, Seal S, Weisbart E, White B, Carpenter AE, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 PMID 41903531 AXIOM BIO
- [4] Seal S, Dee W, Shah A, Cerisier N, Zhang A, Miglietta E, **Titterton K**, Cabrera AA, Boiko D, Beatson A, Slabaugh G, Taboureau O, Carreras Puigvert J, Singh S, Spjuth O, Bender A, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5 PMID 41651839 · PMC12988037 AXIOM BIO
- [5] Wu JW, **Titterton K**, Sebastian R, Romanul N, Yanamandra K, Mastis B, Chang R, Khasnavis S, Kwon T, Chai Y, Manos JD, Preiss CN, Hu M, Welker AM, Liao F, Dellovade T, Karran E, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 PMID 41135680 · PMC12670552 ABBVIE
- [6] Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, Zeng M, Park D, Niederst E, Wei D, Feng G, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w PMID 39521775 · PMC11550395 MIT
- [7] Manos JD, Preiss CN, Venkat N, Tamm J, Reinhardt P, Kwon T, Wu J, Winter AD, Jahn TR, Yanamandra K, **Titterton K**, Karran E, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 PMID 35072001 · PMC8761709 ABBVIE
- [8] Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, Szabo AL, Savechenkov PY, Kent DE, Cohen JB, Forman SA, Miller KW, Buzik KS. Inhibitable photolabeling by neurosteroid diazine analog in the β 3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 PMID 30544077 · PMC6912599 MGH

CONFERENCE PRESENTATIONS & POSTERS

- [1] **Titterton K**, Boiko D, Cabrera AA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. "Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure." *Society of Toxicology (SOT) Annual Meeting*, 2025.
- [2] **Titterton K**, et al. "Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation." *Society for Neuroscience (SfN) Global Connectome Virtual Meeting*, January 2021.
- [3] **Titterton K**, et al. "High content analysis (HCA) of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro." *SBI2 High Content Meeting*, September 2019, Harvard Medical School, Boston, MA.

Earlier research presentations

- "Investigating RNAi as a potential therapeutic for Huntington's Disease in a transgenic mHTT145Q zebra finch (*Taeniopygia guttata*) model." Colgate University Neuroscience Thesis Poster Presentations, 2017.
- "Atopy and helminth infection in Ethiopian schoolchildren." Colgate Summer Research Poster Symposium, 2016.
- "PDE2 regulation of β -arrestin signaling." Systems Biology Center New York (SBCNY) Summer Trainee Poster Symposium, Icahn School of Medicine at Mount Sinai, 2015.